UNIVERSIDAD PERUANA LOS ANDES

FACULTAD DE INGENIERÍA
ESCUELA PROFESIONAL DE INGENIERÍA
DE SISTEMAS Y COMPUTACIÓN

SEMANA 03: PROGRAMACIÓN ORIENTADA A OBJETOS

TALLER DE DESARROLLO

DE APLICACIONES I

SAFORAS GOMEZ Matias Eduardo

HUANCAYO

2026

Ejercicio 01

1. Enunciado del proyecto

Una tienda ha puesto en oferta la venta de un producto ofreciendo un porcentaje de descuento sobre el importe de la compra de acuerdo con la siguiente tabla:

Docenas adquiridas	Descuento
≥ 10	20%
< 10	10%

Adicionalmente, la tienda obsequia lapiceros de acuerdo con a la siguiente tabla:

Importe a pagar	Lapiceros
≥ 200	2 por cada docena
< 200	0

Dado el precio de la docena y la cantidad de docenas adquiridas, diseñe un programa que determine el importe de la compra, el importe del descuento, el importe a pagar y la cantidad de lapiceros de obsequio.

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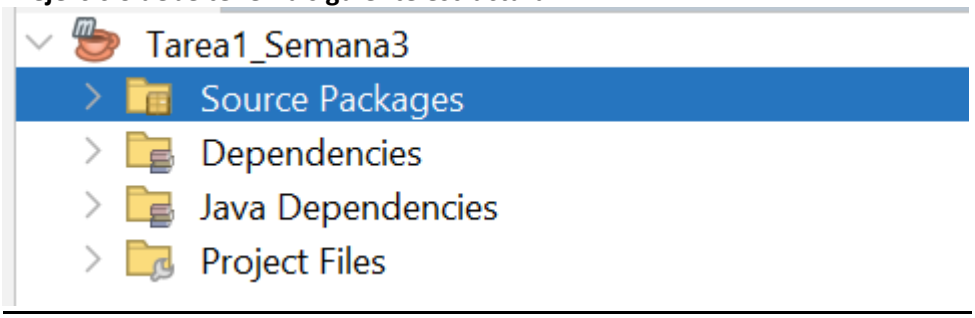
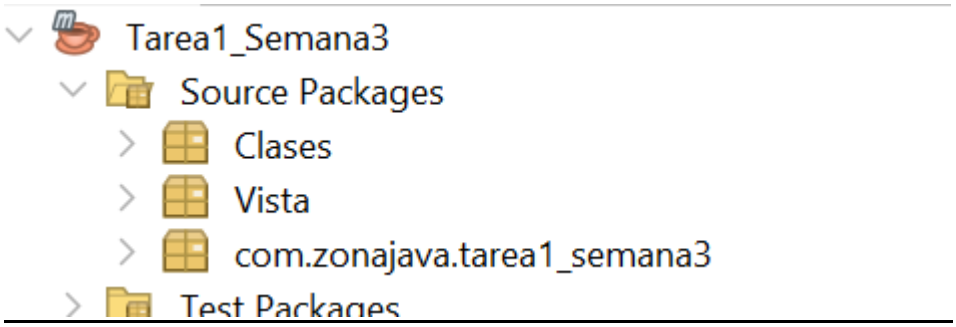
2. Solución

Nombre:	**R1: OfertaTienda**
Resumen:	**Calcular el importe final a pagar por la compra de docenas de un producto.**
Datos de Entrada:	**Precio de la docena del producto, numero de docenas adquiridas.**
Datos de Salida:	**Importe de la compra, Importe del descuento, Importe a pagar, Cantidad de lapiceros de obsequio.**

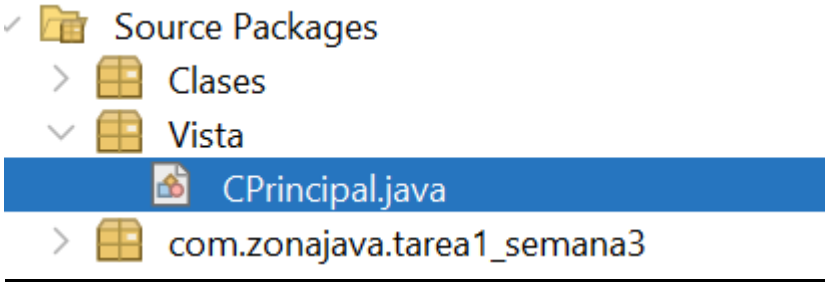
3. El ejercicio debe tener la siguiente estructura



4. **Desplegar la carpeta Source Package y crear un paquete con el nombre VISTA**



5. **Cree en el paquete VISTA, una clase JAVA CLASS con el nombre CPrincipal.java**



6. **Agregue el siguiente código**

```java
package Vista;

import Clases.Tienda;
import java.awt.Color;
import java.awt.Dimension;
import java.awt.event.KeyEvent;
import javax.swing.JOptionPane;

public class CPrincipal extends javax.swing.JFrame {

    Tienda tnd = new Tienda();

    public CPrincipal() {
        initComponents();
        formulario();
    }

    // — Configuración del JFrame ————————————————————————
    private void formulario() {
        this.setTitle("Oferta de Productos - Tienda");
        this.setLocationRelativeTo(null);
        this.setVisible(true);
        this.setSize(new Dimension(400, 185));
        this.setResizable(false);
        this.getContentPane().setBackground(new Color(255, 255, 255));
        this.panelIngreso.setBackground(new Color(255, 255, 255));
        this.panelOperaciones.setBackground(new Color(255, 255, 255));
        this.panelResultado.setBackground(new Color(255, 255, 255));
    }

    // — Estado inicial de paneles y botones ————————————————
    private void valIniciales() {
        this.panelIngreso.setVisible(true);
        this.panelOperaciones.setVisible(true);
        this.panelResultado.setVisible(false);
        this.btnCalcular.setEnabled(false);
        this.btnNuevo.setEnabled(false);
        this.btnSalir.setEnabled(true);
    }

    // — initComponents generado manualmente ————————————————
    @SuppressWarnings("unchecked")
    private void initComponents() {

        panelIngreso      = new javax.swing.JPanel();
        panelOperaciones  = new javax.swing.JPanel();
```

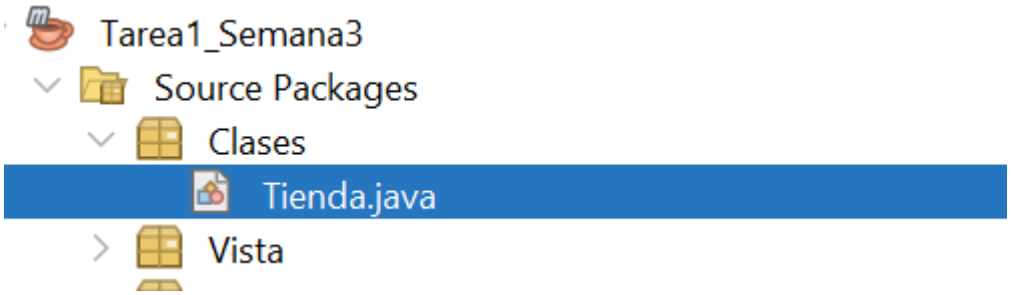
```java
        panelResultado     = new javax.swing.JPanel();

        lblPrecio          = new javax.swing.JLabel();
        lblDocenas         = new javax.swing.JLabel();
        txtPrecio          = new javax.swing.JTextField();
        txtDocenas         = new javax.swing.JTextField();

        btnCalcular        = new javax.swing.JButton();
        btnNuevo           = new javax.swing.JButton();
        btnSalir           = new javax.swing.JButton();

        lblImporteCompra     = new javax.swing.JLabel();
        lblImporteDescuento = new javax.swing.JLabel();
        lblImportePagar      = new javax.swing.JLabel();
        lblLapiceros         = new javax.swing.JLabel();
        txtImporteCompra     = new javax.swing.JTextField();
        txtImporteDescuento = new javax.swing.JTextField();
        txtImportePagar      = new javax.swing.JTextField();
        txtLapiceros         = new javax.swing.JTextField();

        lblResultados = new javax.swing.JLabel();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);

        // — panelIngreso ————————————————————————————————————
        panelIngreso.setBorder(javax.swing.BorderFactory.createEmptyBorder(8, 8, 4, 8));
        panelIngreso.setOpaque(false);

        lblPrecio.setText("Precio por docena (S/.):");
        lblDocenas.setText("Docenas adquiridas:");

        txtPrecio.setPreferredSize(new Dimension(90, 24));
        txtDocenas.setPreferredSize(new Dimension(90, 24));

        txtPrecio.addKeyListener(new java.awt.event.KeyAdapter() {
            public void keyTyped(KeyEvent evt)     { txtPrecioKeyTyped(evt); }
            public void keyPressed(KeyEvent evt)   { txtPrecioKeyPressed(evt); }
        });
        txtDocenas.addKeyListener(new java.awt.event.KeyAdapter() {
            public void keyTyped(KeyEvent evt)     { txtDocenasKeyTyped(evt); }
            public void keyPressed(KeyEvent evt)   { txtDocenasKeyPressed(evt); }
        });

        javax.swing.GroupLayout piLayout = new javax.swing.GroupLayout(panelIngreso);
        panelIngreso.setLayout(piLayout);
        piLayout.setHorizontalGroup(
            piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(piLayout.createSequentialGroup()
                .addGroup(piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addComponent(lblPrecio)
                    .addComponent(lblDocenas))
                .addGap(12)
                .addGroup(piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addComponent(txtPrecio, javax.swing.GroupLayout.PREFERRED_SIZE, 90, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(txtDocenas, javax.swing.GroupLayout.PREFERRED_SIZE, 90, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
        );
        piLayout.setVerticalGroup(
            piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(piLayout.createSequentialGroup()
                .addGroup(piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(lblPrecio)
                    .addComponent(txtPrecio, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addGroup(piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(lblDocenas)
                    .addComponent(txtDocenas, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)))
        );

        // — panelOperaciones ————————————————————————————————
        panelOperaciones.setBorder(javax.swing.BorderFactory.createTitledBorder(
            javax.swing.BorderFactory.createLineBorder(new Color(200, 200, 200)),
            "Operaciones",
            javax.swing.border.TitledBorder.CENTER,
            javax.swing.border.TitledBorder.TOP,
            null, new Color(192, 0, 0)));
        panelOperaciones.setOpaque(false);

        btnCalcular.setText("Calcular");
        btnCalcular.setBackground(new Color(0, 112, 192));
        btnCalcular.setForeground(Color.WHITE);
        btnCalcular.setFocusPainted(false);
        btnCalcular.setMnemonic('C');
        btnCalcular.addActionListener(e -> btnCalcularActionPerformed(e));

        btnNuevo.setText("Nuevo");
        btnNuevo.setBackground(new Color(0, 112, 192));
        btnNuevo.setForeground(Color.WHITE);
        btnNuevo.setFocusPainted(false);
        btnNuevo.setMnemonic('N');
        btnNuevo.addActionListener(e -> btnNuevoActionPerformed(e));

        btnSalir.setText("Salir");
        btnSalir.setBackground(new Color(192, 0, 0));
        btnSalir.setForeground(Color.WHITE);
        btnSalir.setFocusPainted(false);
        btnSalir.setMnemonic('S');
        btnSalir.addActionListener(e -> btnSalirActionPerformed(e));

        javax.swing.GroupLayout poLayout = new javax.swing.GroupLayout(panelOperaciones);
        panelOperaciones.setLayout(poLayout);
```

```java
148         poLayout.setHorizontalGroup(
149             poLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
150             .addGroup(poLayout.createSequentialGroup()
151                 .addContainerGap()
152                 .addComponent(btnCalcular)
153                 .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
154                 .addComponent(btnNuevo)
155                 .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
156                 .addComponent(btnSalir)
157                 .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
158         );
159         poLayout.setVerticalGroup(
160             poLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
161             .addGroup(poLayout.createSequentialGroup()
162                 .addContainerGap()
163                 .addGroup(poLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
164                     .addComponent(btnCalcular)
165                     .addComponent(btnNuevo)
166                     .addComponent(btnSalir))
167                 .addContainerGap())
168         );
169 
170         // -- panelResultado ----------------------------------------
171         panelResultado.setBorder(javax.swing.BorderFactory.createEmptyBorder(4, 8, 8, 8));
172         panelResultado.setOpaque(false);
173 
174         lblResultados.setText("Resultados:");
175         lblResultados.setForeground(new Color(192, 0, 0));
176         lblResultados.setFont(lblResultados.getFont().deriveFont(java.awt.Font.BOLD));
177 
178         lblImporteCompra.setText("Importe de compra (S/.):");
179         lblImporteDescuento.setText("Importe descuento (S/.):");
180         lblImportePagar.setText("Importe a pagar (S/.):");
181         lblLapiceros.setText("Lapiceros de obsequio:");
182 
183         int wRes = 90;
184         txtImporteCompra.setPreferredSize(new Dimension(wRes, 24));
185         txtImporteDescuento.setPreferredSize(new Dimension(wRes, 24));
186         txtImportePagar.setPreferredSize(new Dimension(wRes, 24));
187         txtLapiceros.setPreferredSize(new Dimension(wRes, 24));
188 
189         txtImporteCompra.setEditable(false);
190         txtImporteDescuento.setEditable(false);
191         txtImportePagar.setEditable(false);
192         txtLapiceros.setEditable(false);
193 
194         javax.swing.GroupLayout prLayout = new javax.swing.GroupLayout(panelResultado);
195         panelResultado.setLayout(prLayout);
196         prLayout.setHorizontalGroup(
197             prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
198             .addComponent(lblResultados)
199             .addGroup(prLayout.createSequentialGroup()
200                 .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
201                     .addComponent(lblImporteCompra)
202                     .addComponent(lblImporteDescuento)
203                     .addComponent(lblImportePagar)
204                     .addComponent(lblLapiceros))
205                 .addGap(12)
206                 .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
207                     .addComponent(txtImporteCompra,    javax.swing.GroupLayout.PREFERRED_SIZE, wRes, javax.swing.GroupLayout.PREFERRED_SIZE)
208                     .addComponent(txtImporteDescuento, javax.swing.GroupLayout.PREFERRED_SIZE, wRes, javax.swing.GroupLayout.PREFERRED_SIZE)
209                     .addComponent(txtImportePagar,     javax.swing.GroupLayout.PREFERRED_SIZE, wRes, javax.swing.GroupLayout.PREFERRED_SIZE)
210                     .addComponent(txtLapiceros,        javax.swing.GroupLayout.PREFERRED_SIZE, wRes, javax.swing.GroupLayout.PREFERRED_SIZE))
211                 .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
212         );
213         prLayout.setVerticalGroup(
214             prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
215             .addGroup(prLayout.createSequentialGroup()
216                 .addComponent(lblResultados)
217                 .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
218                 .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
219                     .addComponent(lblImporteCompra)
220                     .addComponent(txtImporteCompra, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
221                 .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
222                 .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
223                     .addComponent(lblImporteDescuento)
224                     .addComponent(txtImporteDescuento, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
225                 .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
226                 .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
227                     .addComponent(lblImportePagar)
228                     .addComponent(txtImportePagar, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
229                 .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
230                 .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
231                     .addComponent(lblLapiceros)
232                     .addComponent(txtLapiceros, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)))
233         );
234 
235         // -- Layout principal ----------------------------------------
236         javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
237         getContentPane().setLayout(layout);
238         layout.setHorizontalGroup(
239             layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
240             .addGroup(layout.createSequentialGroup()
241                 .addContainerGap()
242                 .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
243                     .addComponent(panelIngreso,     javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
244                     .addComponent(panelOperaciones, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
245                     .addComponent(panelResultado,   javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
246                 .addContainerGap())
247         );
248         layout.setVerticalGroup(
249             layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
250             .addGroup(layout.createSequentialGroup()
251                 .addContainerGap()
252                 .addComponent(panelIngreso,     javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
253                 .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
254                 .addComponent(panelOperaciones, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
255                 .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
256                 .addComponent(panelResultado,   javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
257                 .addContainerGap())
258         );
```

```java
        addWindowListener(new java.awt.event.WindowAdapter() {
            public void windowActivated(java.awt.event.WindowEvent e) {
                formWindowActivated(e);
            }
        });

        pack();
    }

    // -- Eventos de botones ----------------------------------------
    private void btnCalcularActionPerformed(java.awt.event.ActionEvent evt) {
        if (txtPrecio.getText().isEmpty() || txtDocenas.getText().isEmpty()) {
            JOptionPane.showMessageDialog(this, "Ingrese un Valor");
        } else {
            float precio  = Float.parseFloat(txtPrecio.getText().trim());
            int   docenas = Integer.parseInt(txtDocenas.getText().trim());

            tnd.setPrecioPorDocena(precio);
            tnd.setDozenasAdquiridas(docenas);

            float importeCompra    = tnd.mostrarImporteCompra();
            float importeDescuento = tnd.mostrarImporteDescuento();
            float importePagar     = tnd.mostrarImportePagar();
            int   lapiceros        = tnd.mostrarLapiceros();

            txtImporteCompra.setText(String.valueOf(importeCompra));
            txtImporteDescuento.setText(String.valueOf(importeDescuento));
            txtImportePagar.setText(String.valueOf(importePagar));
            txtLapiceros.setText(String.valueOf(lapiceros));

            this.setSize(new Dimension(400, 360));
            this.panelResultado.setVisible(true);
            this.btnNuevo.setEnabled(true);
            this.panelResultado.setBackground(new Color(255, 255, 255));
        }
        this.txtPrecio.requestFocus();
        this.btnCalcular.setEnabled(false);
    }

    private void btnNuevoActionPerformed(java.awt.event.ActionEvent evt) {
        this.txtPrecio.setText("");
        this.txtDocenas.setText("");
        this.txtImporteCompra.setText("");
        this.txtImporteDescuento.setText("");
        this.txtImportePagar.setText("");
        this.txtLapiceros.setText("");
        this.setSize(new Dimension(400, 185));
        this.panelResultado.setVisible(false);
        this.txtPrecio.requestFocus();
        this.btnNuevo.setEnabled(false);
        this.btnCalcular.setEnabled(false);
    }

    private void btnSalirActionPerformed(java.awt.event.ActionEvent evt) {
```

```
314            int r = JOptionPane.showOptionDialog(this, "¿Está seguro de salir...?",
315                    "Tienda - Oferta",
316                    JOptionPane.YES_NO_OPTION,
317                    JOptionPane.QUESTION_MESSAGE,
318                    null,
319                    new Object[]{"Si Salgo", "No Salgo"},
320                    "No Salgo");
321            if (r == 0) {
322                System.exit(0);
323            }
324            this.txtPrecio.requestFocus();
325        }
326
327        // -- Eventos de teclado ----------------------------------------
328        private void txtPrecioKeyTyped(KeyEvent evt) {
329            this.btnCalcular.setEnabled(true);
330            char c = evt.getKeyChar();
331            // permite dígitos y punto decimal
332            if (!Character.isDigit(c) && c != '.') {
333                evt.consume();
334            }
335        }
336
337        private void txtPrecioKeyPressed(KeyEvent evt) {
338            if (evt.getKeyCode() == KeyEvent.VK_ENTER) {
339                btnCalcular.doClick();
340            }
341        }
342
343        private void txtDocenasKeyTyped(KeyEvent evt) {
344            this.btnCalcular.setEnabled(true);
345            char c = evt.getKeyChar();
346            if (!Character.isDigit(c)) {
347                evt.consume();
348            }
349        }
350
351        private void txtDocenasKeyPressed(KeyEvent evt) {
352            if (evt.getKeyCode() == KeyEvent.VK_ENTER) {
353                btnCalcular.doClick();
354            }
355        }
356
357        private void formWindowActivated(java.awt.event.WindowEvent evt) {
358            valIniciales();
359        }
360
361        // -- Variables de declaración ----------------------------------
362        private javax.swing.JButton    btnCalcular;
363        private javax.swing.JButton    btnNuevo;
364        private javax.swing.JButton    btnSalir;
365        private javax.swing.JLabel     lblPrecio;
366        private javax.swing.JLabel     lblDocenas;
367        private javax.swing.JLabel     lblResultados;
368        private javax.swing.JLabel     lblImporteCompra;
369        private javax.swing.JLabel     lblImporteDescuento;
370        private javax.swing.JLabel     lblImportePagar;
371        private javax.swing.JLabel     lblLapiceros;
372        private javax.swing.JPanel     panelIngreso;
373        private javax.swing.JPanel     panelOperaciones;
374        private javax.swing.JPanel     panelResultado;
375        private javax.swing.JTextField txtPrecio;
376        private javax.swing.JTextField txtDocenas;
377        private javax.swing.JTextField txtImporteCompra;
378        private javax.swing.JTextField txtImporteDescuento;
379        private javax.swing.JTextField txtImportePagar;
380        private javax.swing.JTextField txtLapiceros;
381    }
```

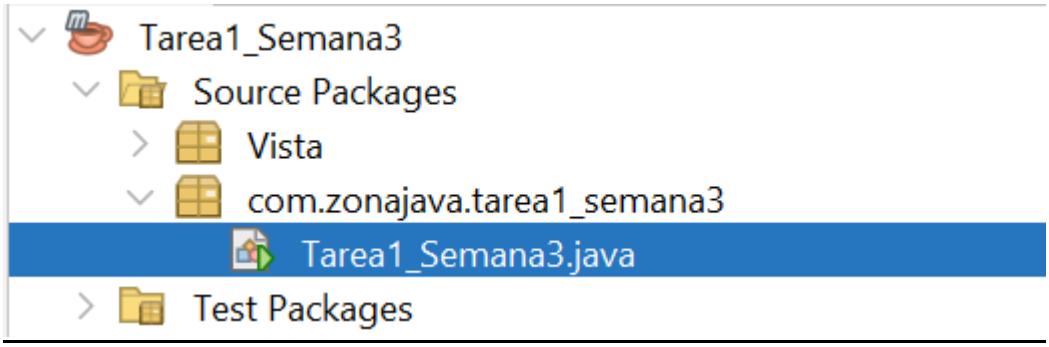
7. **Cree en el paquete CLASES, una clase JAVA CLASS con el nombre Tienda.java**



8. **Agregue el siguiente código**

```java
package Clases;

public class Tienda {

    private float precioPorDocena;
    private int   dozenasAdquiridas;

    private float importeCompra;
    private float importeDescuento;
    private float importePagar;
    private int   lapiceros;

    public Tienda() {
    }

    public Tienda(float precioPorDocena, int dozenasAdquiridas) {
        this.precioPorDocena   = precioPorDocena;
        this.dozenasAdquiridas = dozenasAdquiridas;
    }

    // — Getters y Setters ————————————————————————————
    public float getPrecioPorDocena()            { return precioPorDocena; }
    public void  setPrecioPorDocena(float p)     { this.precioPorDocena = p; }

    public int  getDozenasAdquiridas()           { return dozenasAdquiridas; }
    public void setDozenasAdquiridas(int d)      { this.dozenasAdquiridas = d; }

    // — Métodos de cálculo ————————————————————————————
    public float calcularImporteCompra() {
        importeCompra = precioPorDocena * dozenasAdquiridas;
        return importeCompra;
    }

    public float calcularImporteDescuento() {
        if (dozenasAdquiridas >= 10) {
            importeDescuento = 0.20f * importeCompra;
        } else {
            importeDescuento = 0.10f * importeCompra;
        }
        return importeDescuento;
    }

    public float calcularImportePagar() {
        importePagar = importeCompra - importeDescuento;
        return importePagar;
    }

    public int calcularLapiceros() {
        if (importePagar >= 200) {
            lapiceros = 2 * dozenasAdquiridas;
        } else {
            lapiceros = 0;
        }
        return lapiceros;
    }

    // — Getters de resultados ————————————————————————
    public float mostrarImporteCompra()     { return calcularImporteCompra(); }
    public float mostrarImporteDescuento()  { return calcularImporteDescuento(); }
    public float mostrarImportePagar()      { return calcularImportePagar(); }
    public int   mostrarLapiceros()         { return calcularLapiceros(); }
}
```

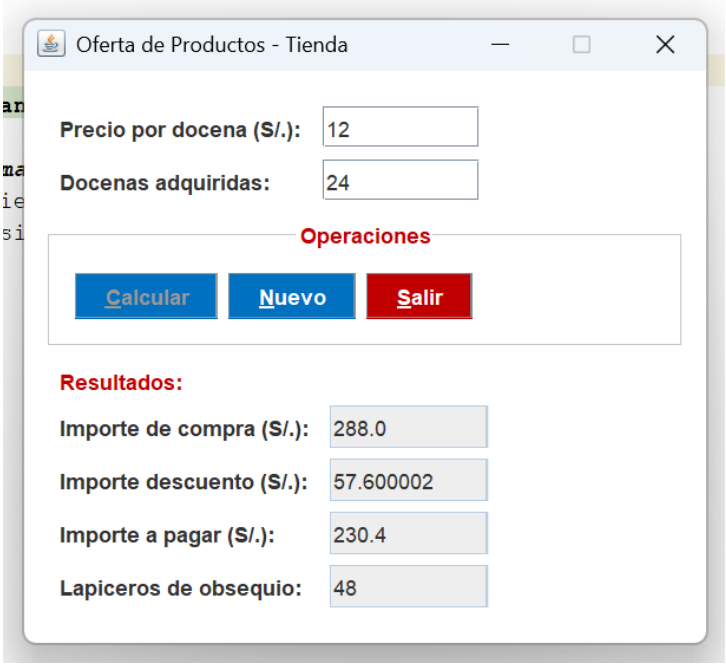
9. **Abrir la clase Tarea1_Semana3.java del paquete: com.zonajava.tarea1_semana3**



10. **Modificar con el siguiente código**

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 */
package com.zonajava.tarea1_semana3;

import Vista.ImporteT;

/**
 *
 * @author Matias
 */
public class Tarea1_Semana3 {

    public static void main(String[] args) {
        ImporteT OfertaTienda = new ImporteT();
        OfertaTienda.show();
    }
}

```

11. **Ejecución de la Aplicación**



Ejercicio 02

12. Enunciado del Proyecto

Una empresa de transportes cubre la ruta Lima – Huancayo en dos turnos: mañana y noche. Los precios de los pasajes se dan en la siguiente tabla:

Turno	Precio del pasaje
Mañana	S/. 37.5
Noche	S/. 37.5

Como oferta especial, la empresa aplica un porcentaje de descuento sobre el importe de la compra de acuerdo con a la siguiente tabla:

Cantidad de pasajes	Descuento
≥ 15	8%
< 15	5%

Adicionalmente, la empresa obsequia caramelos de acuerdo con la siguiente tabla.

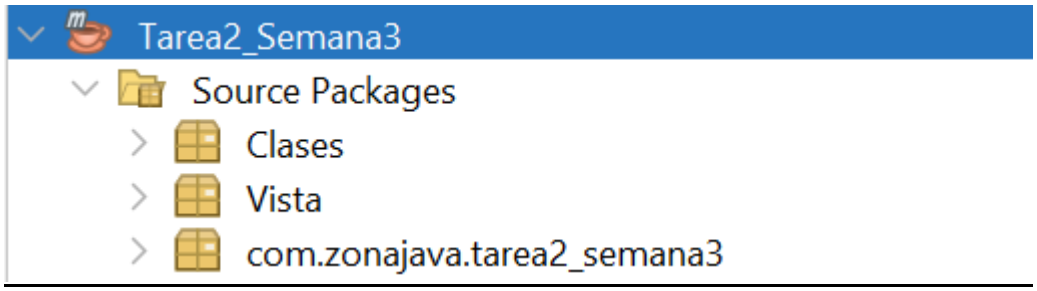
Importe por pagar	Descuento
> 200	2 por cada boleto
≤ 200	0

Dado el turno y la cantidad de pasajes adquiridos por un cliente, diseñe un programa que determine el importe de la compra, el importe del descuento, el importe a pagar y la cantidad de caramelos de obsequio.

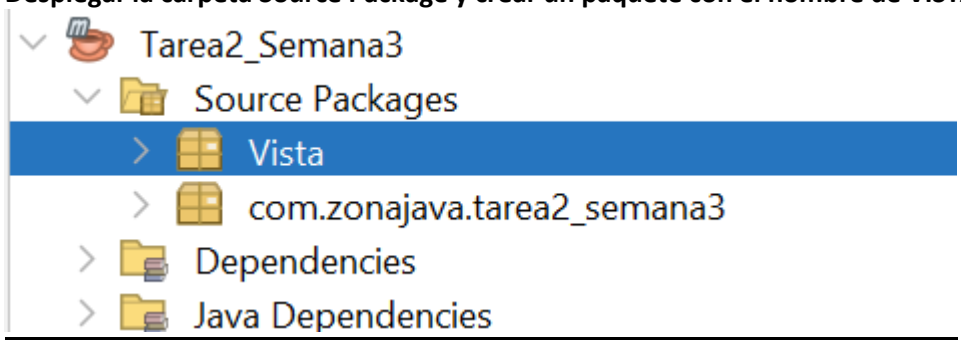
13. **Solución**

Nombre:	**R1: VentaPasajes**
Resumen:	**Calcular el importe a pagar por la compra de pasajes de bus según el turno.**
Datos de Entrada:	**Turno (mañana/noche), Cantidad número de pasajes.**
Datos de Salida:	**Importe de la compra, descuento, pagar y cantidad de caramelos de obsequio.**

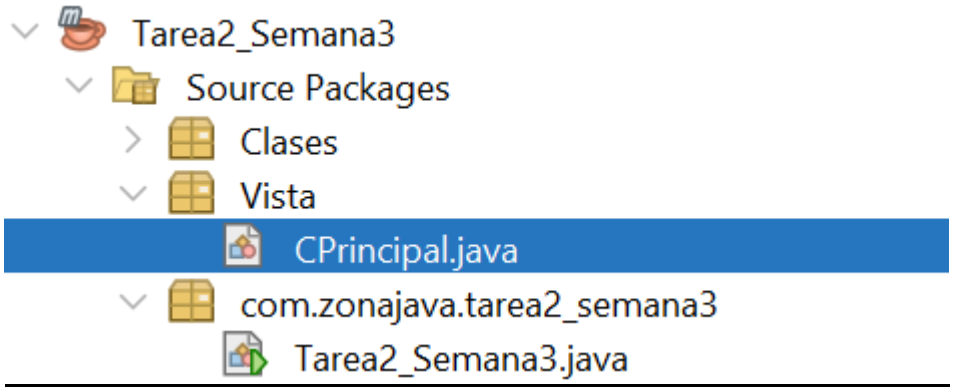
14. **El proyecto resultará con la siguiente estructura**



15. **Desplegar la carpeta Source Package y crear un paquete con el nombre de VISTA**



16. **Cree en el paquete VISTA, una clase JAVA CLASS con el nombre de CPrincipal.java**



17. Agregue el siguiente código:

```java
package Vista;

import Clases.Transporte;
import java.awt.Color;
import java.awt.Dimension;
import java.awt.event.KeyEvent;
import javax.swing.DefaultListCellRenderer;
import javax.swing.JOptionPane;

public class CPrincipal extends javax.swing.JFrame {

    Transporte trp = new Transporte();

    public CPrincipal() {
        initComponents();
        formulario();
    }

    // — Configuración del JFrame ————————————————————————
    private void formulario() {
        this.setTitle("Empresa de Transportes Lima - Huancayo");
        this.setLocationRelativeTo(null);
        this.setVisible(true);
        this.setSize(new Dimension(420, 195));
        this.setResizable(false);
        this.getContentPane().setBackground(new Color(255, 255, 255));
        this.panelIngreso.setBackground(new Color(255, 255, 255));
        this.panelOperaciones.setBackground(new Color(255, 255, 255));
        this.panelResultado.setBackground(new Color(255, 255, 255));

        // Inicializar ComboBox turno
        DefaultListCellRenderer cr = new DefaultListCellRenderer();
        cr.setHorizontalAlignment(DefaultListCellRenderer.CENTER);
        cboTurno.setRenderer(cr);
        cboTurno.addItem("Seleccione");
        cboTurno.addItem("Mañana  (S/. 37.5)");
        cboTurno.addItem("Noche   (S/. 37.5)");
        cboTurno.requestFocus();
    }

    // — Estado inicial ————————————————————————————————
    private void valIniciales() {
        this.panelIngreso.setVisible(true);
        this.panelOperaciones.setVisible(true);
        this.panelResultado.setVisible(false);
        this.btnCalcular.setEnabled(false);
        this.btnNuevo.setEnabled(false);
        this.btnSalir.setEnabled(true);
    }

    // — initComponents ————————————————————————————————
    @SuppressWarnings("unchecked")
    private void initComponents() {

        panelIngreso     = new javax.swing.JPanel();
        panelOperaciones = new javax.swing.JPanel();
        panelResultado   = new javax.swing.JPanel();

        lblTurno     = new javax.swing.JLabel();
        lblPasajes   = new javax.swing.JLabel();
        cboTurno     = new javax.swing.JComboBox<>();
        txtPasajes   = new javax.swing.JTextField();

        btnCalcular = new javax.swing.JButton();
        btnNuevo    = new javax.swing.JButton();
        btnSalir    = new javax.swing.JButton();

        lblResultados       = new javax.swing.JLabel();
        lblImporteCompra    = new javax.swing.JLabel();
        lblImporteDescuento = new javax.swing.JLabel();
        lblImportePagar     = new javax.swing.JLabel();
        lblCaramelos        = new javax.swing.JLabel();
        txtImporteCompra    = new javax.swing.JTextField();
        txtImporteDescuento = new javax.swing.JTextField();
        txtImportePagar     = new javax.swing.JTextField();
        txtCaramelos        = new javax.swing.JTextField();

        setDefaultCloseOperation(javax.swing.WindowConstants.EXIT_ON_CLOSE);

        // — panelIngreso ——————————————————————————————
        panelIngreso.setBorder(javax.swing.BorderFactory.createEmptyBorder(8, 8, 4, 8));
        panelIngreso.setOpaque(false);

        lblTurno.setText("Turno:");
        lblPasajes.setText("Cantidad de pasajes:");

        cboTurno.setPreferredSize(new Dimension(160, 24));
        txtPasajes.setPreferredSize(new Dimension(90, 24));

        cboTurno.addActionListener(e -> cboTurnoChanged());
        txtPasajes.addKeyListener(new java.awt.event.KeyAdapter() {
            public void keyTyped(KeyEvent evt)   { txtPasajesKeyTyped(evt); }
            public void keyPressed(KeyEvent evt) { txtPasajesKeyPressed(evt); }
        });

        javax.swing.GroupLayout piLayout = new javax.swing.GroupLayout(panelIngreso);
        panelIngreso.setLayout(piLayout);
        piLayout.setHorizontalGroup(
            piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(piLayout.createSequentialGroup()
                .addGroup(piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addComponent(lblTurno)
                    .addComponent(lblPasajes))
                .addGap(18)
                .addGroup(piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addComponent(cboTurno,   javax.swing.GroupLayout.PREFERRED_SIZE, 160, javax.swing.GroupLayout.PREFERRED_SIZE)
                    .addComponent(txtPasajes, javax.swing.GroupLayout.PREFERRED_SIZE, 90,  javax.swing.GroupLayout.PREFERRED_SIZE))
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
        );
        piLayout.setVerticalGroup(
```

```java
        piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(piLayout.createSequentialGroup()
            .addGroup(piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(lblTurno)
                .addComponent(cboTurno, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addGroup(piLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(lblPasajes)
                .addComponent(txtPasajes, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)))
    );

    // ── panelOperaciones ──────────────────────────────────────────
    panelOperaciones.setBorder(javax.swing.BorderFactory.createTitledBorder(
        javax.swing.BorderFactory.createLineBorder(new Color(200, 200, 200)),
        "Operaciones",
        javax.swing.border.TitledBorder.CENTER,
        javax.swing.border.TitledBorder.TOP,
        null, new Color(192, 0, 0)));
    panelOperaciones.setOpaque(false);

    btnCalcular.setText("Calcular");
    btnCalcular.setBackground(new Color(0, 112, 192));
    btnCalcular.setForeground(Color.WHITE);
    btnCalcular.setFocusPainted(false);
    btnCalcular.setMnemonic('C');
    btnCalcular.addActionListener(e -> btnCalcularActionPerformed());

    btnNuevo.setText("Nuevo");
    btnNuevo.setBackground(new Color(0, 112, 192));
    btnNuevo.setForeground(Color.WHITE);
    btnNuevo.setFocusPainted(false);
    btnNuevo.setMnemonic('N');
    btnNuevo.addActionListener(e -> btnNuevoActionPerformed());

    btnSalir.setText("Salir");
    btnSalir.setBackground(new Color(192, 0, 0));
    btnSalir.setForeground(Color.WHITE);
    btnSalir.setFocusPainted(false);
    btnSalir.setMnemonic('S');
    btnSalir.addActionListener(e -> btnSalirActionPerformed());

    javax.swing.GroupLayout poLayout = new javax.swing.GroupLayout(panelOperaciones);
    panelOperaciones.setLayout(poLayout);
    poLayout.setHorizontalGroup(
        poLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(poLayout.createSequentialGroup()
            .addContainerGap()
            .addComponent(btnCalcular)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(btnNuevo)
            .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
            .addComponent(btnSalir)
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
    );
    poLayout.setVerticalGroup(
```

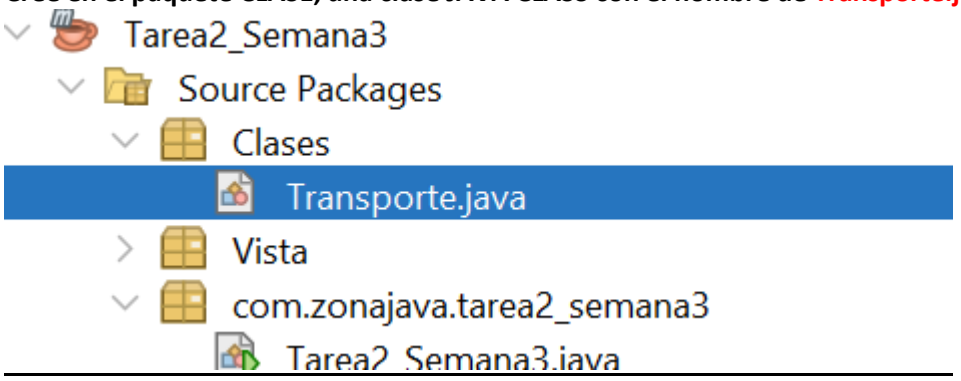

```java
    poLayout.setVerticalGroup(
        poLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(poLayout.createSequentialGroup()
            .addContainerGap()
            .addGroup(poLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                .addComponent(btnCalcular)
                .addComponent(btnNuevo)
                .addComponent(btnSalir))
            .addContainerGap())
    );
```


```java
    // ── panelResultado ────────────────────────────────────────────
    panelResultado.setBorder(javax.swing.BorderFactory.createEmptyBorder(4, 8, 8, 8));
    panelResultado.setOpaque(false);

    lblResultados.setText("Resultados:");
    lblResultados.setForeground(new Color(192, 0, 0));
    lblResultados.setFont(lblResultados.getFont().deriveFont(java.awt.Font.BOLD));

    lblImporteCompra.setText("Importe de compra (S/.):");
    lblImporteDescuento.setText("Importe descuento (S/.):");
    lblImportePagar.setText("Importe a pagar (S/.):");
    lblCaramelos.setText("Caramelos de obsequio:");

    int wRes = 90;
    txtImporteCompra.setPreferredSize(new Dimension(wRes, 24));
    txtImporteDescuento.setPreferredSize(new Dimension(wRes, 24));
    txtImportePagar.setPreferredSize(new Dimension(wRes, 24));
    txtCaramelos.setPreferredSize(new Dimension(wRes, 24));

    txtImporteCompra.setEditable(false);
    txtImporteDescuento.setEditable(false);
    txtImportePagar.setEditable(false);
    txtCaramelos.setEditable(false);

    javax.swing.GroupLayout prLayout = new javax.swing.GroupLayout(panelResultado);
    panelResultado.setLayout(prLayout);
    prLayout.setHorizontalGroup(
        prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addComponent(lblResultados)
        .addGroup(prLayout.createSequentialGroup()
            .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(lblImporteCompra)
                .addComponent(lblImporteDescuento)
                .addComponent(lblImportePagar)
                .addComponent(lblCaramelos))
            .addGap(12)
            .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                .addComponent(txtImporteCompra,    javax.swing.GroupLayout.PREFERRED_SIZE, wRes, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(txtImporteDescuento, javax.swing.GroupLayout.PREFERRED_SIZE, wRes, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(txtImportePagar,     javax.swing.GroupLayout.PREFERRED_SIZE, wRes, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addComponent(txtCaramelos,        javax.swing.GroupLayout.PREFERRED_SIZE, wRes, javax.swing.GroupLayout.PREFERRED_SIZE))
            .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
    );
    prLayout.setVerticalGroup(
```

```java
            prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(prLayout.createSequentialGroup()
                .addComponent(lblResultados)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(lblImporteCompra)
                    .addComponent(txtImporteCompra, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(lblImporteDescuento)
                    .addComponent(txtImporteDescuento, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(lblImportePagar)
                    .addComponent(txtImportePagar, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE))
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addGroup(prLayout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
                    .addComponent(lblCaramelos)
                    .addComponent(txtCaramelos, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)))
        );

        // — Layout principal ————————————————————————————————
        javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
        getContentPane().setLayout(layout);
        layout.setHorizontalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addGroup(layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
                    .addComponent(panelIngreso,     javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addComponent(panelOperaciones, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
                    .addComponent(panelResultado,   javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE))
                .addContainerGap())
        );
        layout.setVerticalGroup(
            layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(panelIngreso,     javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(panelOperaciones, javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(panelResultado,   javax.swing.GroupLayout.PREFERRED_SIZE, javax.swing.GroupLayout.DEFAULT_SIZE, javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap())
        );

        addWindowListener(new java.awt.event.WindowAdapter() {
            public void windowActivated(java.awt.event.WindowEvent e) {
                valIniciales();
            }
        });

        pack();
    }

    // — Eventos ————————————————————————————————————————
    private void cboTurnoChanged() {
        // habilitar calcular solo si se eligió turno Y hay pasajes
        verificarHabilitar();
    }

    private void verificarHabilitar() {
        boolean turnoOk   = cboTurno.getSelectedIndex() > 0;
        boolean pasajesOk = !txtPasajes.getText().isEmpty();
        btnCalcular.setEnabled(turnoOk && pasajesOk);
    }

    private void btnCalcularActionPerformed() {
        if (cboTurno.getSelectedIndex() == 0 || txtPasajes.getText().isEmpty()) {
            JOptionPane.showMessageDialog(this, "Ingrese todos los datos", "Advertencia",
                    JOptionPane.WARNING_MESSAGE);
            return;
        }

        int turno    = cboTurno.getSelectedIndex(); // 1=Mañana, 2=Noche
        int pasajes  = Integer.parseInt(txtPasajes.getText().trim());

        trp.setTurno(turno);
        trp.setCantidadPasajes(pasajes);

        float importeCompra    = trp.mostrarImporteCompra();
        float importeDescuento = trp.mostrarImporteDescuento();
        float importePagar     = trp.mostrarImportePagar();
        int   caramelos        = trp.mostrarCaramelos();

        txtImporteCompra.setText(String.format("%.2f", importeCompra));
        txtImporteDescuento.setText(String.format("%.2f", importeDescuento));
        txtImportePagar.setText(String.format("%.2f", importePagar));
        txtCaramelos.setText(String.valueOf(caramelos));

        this.setSize(new Dimension(420, 375));
        this.panelResultado.setVisible(true);
        this.btnNuevo.setEnabled(true);
        this.btnCalcular.setEnabled(false);
        this.txtPasajes.requestFocus();
    }

    private void btnNuevoActionPerformed() {
        this.cboTurno.setSelectedIndex(0);
        this.txtPasajes.setText("");
        this.txtImporteCompra.setText("");
        this.txtImporteDescuento.setText("");
        this.txtImportePagar.setText("");
        this.txtCaramelos.setText("");
        this.setSize(new Dimension(420, 195));
        this.panelResultado.setVisible(false);
        this.btnNuevo.setEnabled(false);
        this.btnCalcular.setEnabled(false);
```

```java
        this.cboTurno.requestFocus();
    }

    private void btnSalirActionPerformed() {
        int r = JOptionPane.showOptionDialog(this,
                "¿Está seguro de salir...?",
                "Transportes Lima - Huancayo",
                JOptionPane.YES_NO_OPTION,
                JOptionPane.QUESTION_MESSAGE,
                null,
                new Object[]{"Si Salgo", "No Salgo"},
                "No Salgo");
        if (r == 0) {
            System.exit(0);
        }
        this.cboTurno.requestFocus();
    }

    private void txtPasajesKeyTyped(KeyEvent evt) {
        char c = evt.getKeyChar();
        if (!Character.isDigit(c)) {
            evt.consume();
        } else {
            verificarHabilitar();
        }
    }

    private void txtPasajesKeyPressed(KeyEvent evt) {
        if (evt.getKeyCode() == KeyEvent.VK_ENTER) {
            btnCalcular.doClick();
        }
    }

    // -- Variables de declaración ----------------------------------------
    private javax.swing.JButton              btnCalcular;
    private javax.swing.JButton              btnNuevo;
    private javax.swing.JButton              btnSalir;
    private javax.swing.JComboBox<String>    cboTurno;
    private javax.swing.JLabel               lblTurno;
    private javax.swing.JLabel               lblPasajes;
    private javax.swing.JLabel               lblResultados;
    private javax.swing.JLabel               lblImporteCompra;
    private javax.swing.JLabel               lblImporteDescuento;
    private javax.swing.JLabel               lblImportePagar;
    private javax.swing.JLabel               lblCaramelos;
    private javax.swing.JPanel               panelIngreso;
    private javax.swing.JPanel               panelOperaciones;
    private javax.swing.JPanel               panelResultado;
    private javax.swing.JTextField           txtPasajes;
    private javax.swing.JTextField           txtImporteCompra;
    private javax.swing.JTextField           txtImporteDescuento;
    private javax.swing.JTextField           txtImportePagar;
    private javax.swing.JTextField           txtCaramelos;
}
```

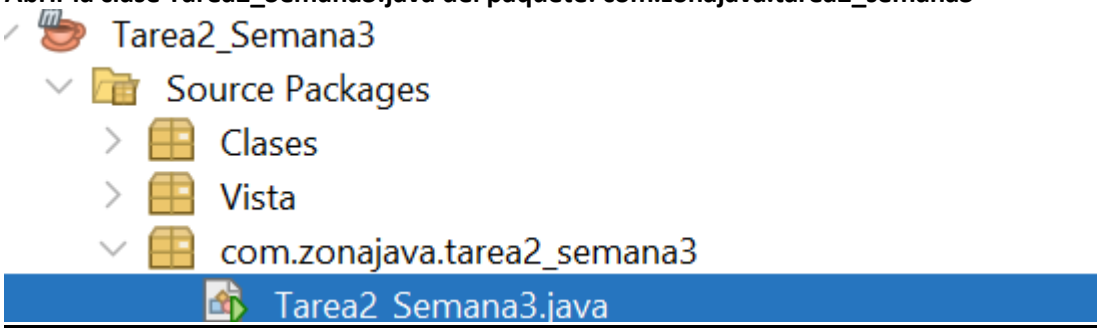
18. Cree en el paquete CLASE, una clase JAVA CLASS con el nombre de Transporte.java



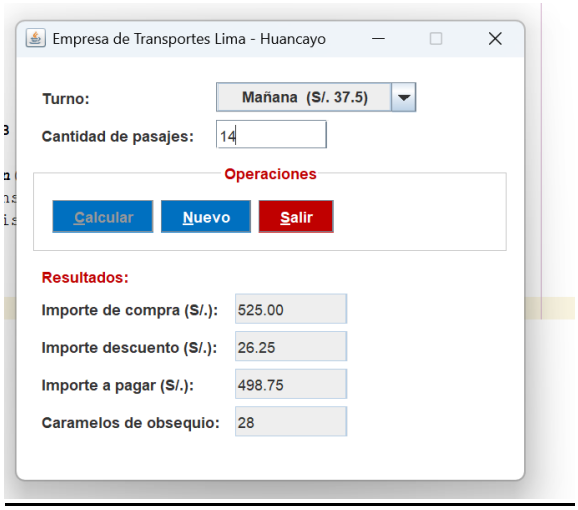
19. Modificar con el siguiente código

```java
package Clases;

public class Transporte {

    private int    turno;            // 1 = Mañana, 2 = Noche
    private int    cantidadPasajes;

    private float  importeCompra;
    private float  importeDescuento;
    private float  importePagar;
    private int    caramelos;

    // Precio único para ambos turnos
    private static final float PRECIO_PASAJE = 37.5f;

    public Transporte() {
    }

    public Transporte(int turno, int cantidadPasajes) {
        this.turno            = turno;
        this.cantidadPasajes = cantidadPasajes;
    }

    // — Getters y Setters ————————————————————————————————
    public int  getTurno()                    { return turno; }
    public void setTurno(int turno)           { this.turno = turno; }

    public int  getCantidadPasajes()            { return cantidadPasajes; }
    public void setCantidadPasajes(int c)       { this.cantidadPasajes = c; }

    // — Métodos de cálculo ————————————————————————————————
    public float calcularImporteCompra() {
        importeCompra = PRECIO_PASAJE * cantidadPasajes;
        return importeCompra;
    }

    public float calcularImporteDescuento() {
        if (cantidadPasajes >= 15) {
            importeDescuento = 0.08f * importeCompra;
        } else {
            importeDescuento = 0.05f * importeCompra;
        }
        return importeDescuento;
    }

    public float calcularImportePagar() {
        importePagar = importeCompra - importeDescuento;
        return importePagar;
    }

    public int calcularCaramelos() {
        if (importePagar > 200) {
            caramelos = 2 * cantidadPasajes;
        } else {
            caramelos = 0;
        }
        return caramelos;
    }

    // — Métodos mostrar (encadenados en orden correcto) ——————————
    public float mostrarImporteCompra() {
        return calcularImporteCompra();
    }

    public float mostrarImporteDescuento() {
        calcularImporteCompra();
        return calcularImporteDescuento();
    }

    public float mostrarImportePagar() {
        calcularImporteCompra();
        calcularImporteDescuento();
        return calcularImportePagar();
    }

    public int mostrarCaramelos() {
        calcularImporteCompra();
        calcularImporteDescuento();
        calcularImportePagar();
        return calcularCaramelos();
    }
}
```

20. Abrir la clase Tarea2_Semana3.java del paquete: com.zonajava.tarea2_semana3



21. Modificar con el siguiente código

```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 */
package com.zonajava.tarea2_semana3;

import Vista.CPrincipal;

/**
 *
 * @author Matias
 */
public class Tarea2_Semana3 {

    public static void main(String[] args) {
        CPrincipal objTransporte = new CPrincipal();
        objTransporte.setVisible(true);
    }
}
```

22. Ejecución de la Aplicación